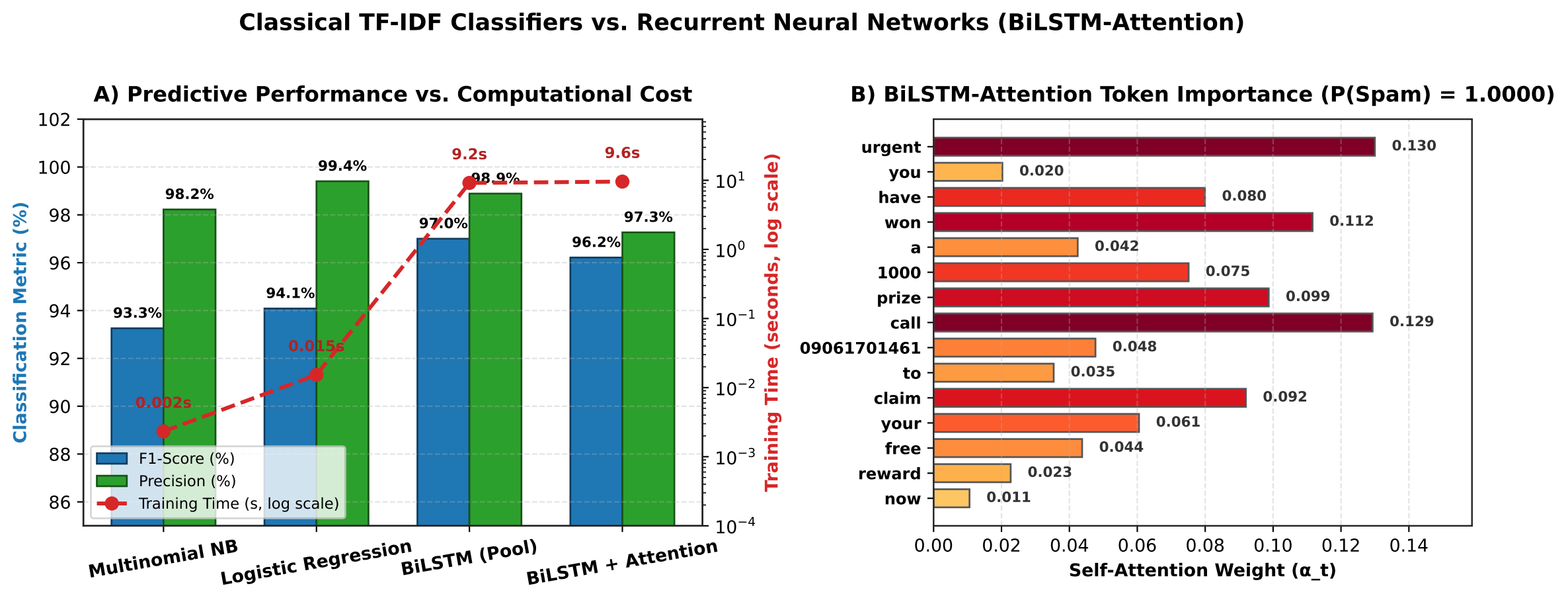
Classical TF-IDF Classifiers vs. Recurrent Neural Networks (BiLSTM-Attention)
A) Predictive Performance vs. Computational Cost
Classification Metric (%)
Training Time (seconds, log scale)
102
100
98
96
94
92
90
88
86
10^{1}
10^{0}
10^{-1}
10^{-2}
10^{-3}
10^{-4}
93.3%
98.2%
94.1%
99.4%
97.0%
98.9%
96.2%
97.3%
0.002s
0.015s
9.2s
9.6s
F1-Score (%)
Precision (%)
Training Time (s, log scale)
Multinomial NB
Logistic Regression
BiLSTM (Pool)
BiLSTM + Attention
B) BiLSTM-Attention Token Importance (P(Spam) = 1.0000)
urgent 0.130
you 0.020
have 0.080
won 0.112
a 0.042
1000 0.075
prize 0.099
call 0.129
09061701461 0.048
to 0.035
claim 0.092
your 0.061
free 0.044
reward 0.023
now 0.011
0.00
0.02
0.04
0.06
0.08
0.10
0.12
0.14
Self-Attention Weight (α_t)